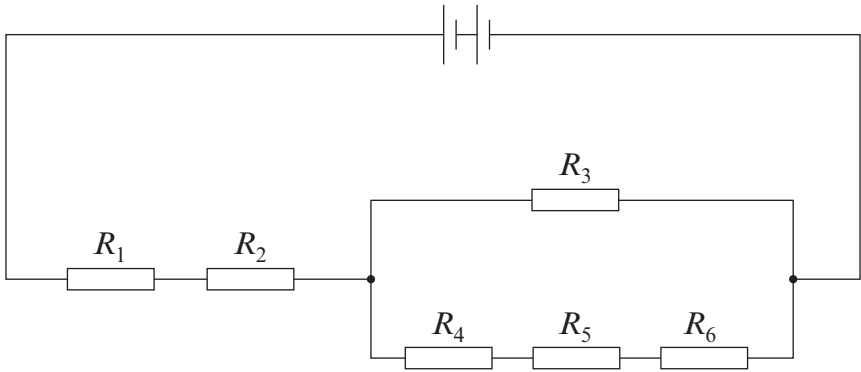


7. The diagram below shows six resistors connected to a battery.



The resistor values are given in the table below.

Resistor	R_1	R_2	R_3	R_4	R_5	R_6
Resistance (Ω)	60	80	300	160	120	170

Use equations on page 2 to answer the following questions:

(a) Calculate the total resistance of the series combination R_4 , R_5 and R_6 . [1]

total resistance of R_4 , R_5 and R_6 = Ω

(b) Calculate the total resistance of R_3 combined in parallel with R_4 , R_5 and R_6 . [3]

total resistance of the parallel combination = Ω

(c) Calculate the total resistance of the circuit. [1]

total resistance of circuit = Ω

5

